

Crank-Nicolson Discretization of Usadel Equation

Simple Example: Two Derivative Terms

Consider the equation:

$$i\hat{\tau}_3 \frac{\partial g}{\partial t_1} + \frac{\partial g}{\partial t_2} i\hat{\tau}_3 = 0, \quad (1)$$

where $g = g(t_1, t_2)$ is a function of two time arguments.

Goal: Evolve g from time t to $t + \Delta t$ in the first argument t_1 , holding $t_2 = t'$ fixed.

Step 1: Integrate over time interval. Integrate the equation from t to $t + \Delta t$ in the t_1 direction:

$$\int_t^{t+\Delta t} dt_1 \left[i\hat{\tau}_3 \frac{\partial g(t_1, t')}{\partial t_1} + \frac{\partial g(t_1, t')}{\partial t'} i\hat{\tau}_3 \right] = 0. \quad (2)$$

Step 2: Forward Euler (explicit scheme). Evaluate the integrand at the *beginning* of the interval ($t_1 = t$):

$$\Delta t \cdot \left[i\hat{\tau}_3 \frac{\partial g(t, t')}{\partial t_1} \Big|_{t_1=t} + \frac{\partial g(t, t')}{\partial t'} i\hat{\tau}_3 \right] = 0. \quad (3)$$

Using finite differences:

$$\frac{\partial g}{\partial t_1} \Big|_{t_1=t} \approx \frac{g(t + \Delta t, t') - g(t, t')}{\Delta t}, \quad (4)$$

$$\frac{\partial g}{\partial t'} \approx \frac{g(t, t' + \Delta t) - g(t, t')}{\Delta t}. \quad (5)$$

Substituting:

$$i\hat{\tau}_3 [g(t + \Delta t, t') - g(t, t')] + [g(t, t' + \Delta t) - g(t, t')] i\hat{\tau}_3 = 0. \quad (6)$$

Solving for the unknown $g(t + \Delta t, t')$:

$$\boxed{i\hat{\tau}_3 g(t + \Delta t, t') = -[g(t, t' + \Delta t) - g(t, t')] i\hat{\tau}_3 + i\hat{\tau}_3 g(t, t').} \quad (7)$$

This is an **explicit update**: $g(t + \Delta t, t')$ can be directly computed from known values at time t .

Step 3: Crank-Nicolson (implicit scheme). Evaluate the integrand at *both* the beginning ($t_1 = t$) and end ($t_1 = t + \Delta t$) of the interval, then average:

$$\begin{aligned} & \frac{\Delta t}{2} \left[i\hat{\tau}_3 \frac{\partial g(t, t')}{\partial t_1} \Big|_{t_1=t} + \frac{\partial g(t, t')}{\partial t'} i\hat{\tau}_3 \right] \\ & + \frac{\Delta t}{2} \left[i\hat{\tau}_3 \frac{\partial g(t + \Delta t, t')}{\partial t_1} \Big|_{t_1=t+\Delta t} + \frac{\partial g(t + \Delta t, t')}{\partial t'} i\hat{\tau}_3 \right] = 0. \end{aligned} \quad (8)$$

Using finite differences:

$$\frac{\partial g}{\partial t_1} \Big|_{t_1=t} \approx \frac{g(t + \Delta t, t') - g(t, t')}{\Delta t}, \quad (9)$$

$$\frac{\partial g(t, t')}{\partial t'} \approx \frac{g(t, t' + \Delta t) - g(t, t')}{\Delta t}, \quad (10)$$

$$\frac{\partial g}{\partial t_1} \Big|_{t_1=t+\Delta t} \approx \frac{g(t + 2\Delta t, t') - g(t + \Delta t, t')}{\Delta t}, \quad (11)$$

$$\frac{\partial g(t + \Delta t, t')}{\partial t'} \approx \frac{g(t + \Delta t, t' + \Delta t) - g(t + \Delta t, t')}{\Delta t}. \quad (12)$$

Substituting and simplifying:

$$\begin{aligned} & \frac{1}{2} \left[i\hat{\tau}_3 (g(t + \Delta t, t') - g(t, t')) + (g(t, t' + \Delta t) - g(t, t')) i\hat{\tau}_3 \right] \\ & + \frac{1}{2} \left[i\hat{\tau}_3 (g(t + 2\Delta t, t') - g(t + \Delta t, t')) + (g(t + \Delta t, t' + \Delta t) - g(t + \Delta t, t')) i\hat{\tau}_3 \right] = 0. \end{aligned} \quad (13)$$

Problem: This involves $g(t + 2\Delta t, t')$, which we don't know yet!

Modified Crank-Nicolson: Use centered difference for ∂_{t_1} derivative only:

$$\frac{\partial g}{\partial t_1} \Big|_{t_1=t+\Delta t/2} \approx \frac{g(t + \Delta t, t') - g(t, t')}{\Delta t}, \quad (14)$$

and average the $\partial_{t'}$ derivative at times t and $t + \Delta t$:

$$\frac{1}{2} \left[\frac{\partial g(t, t')}{\partial t'} + \frac{\partial g(t + \Delta t, t')}{\partial t'} \right] \approx \frac{1}{2} \left[\frac{g(t, t' + \Delta t) - g(t, t')}{\Delta t} + \frac{g(t + \Delta t, t' + \Delta t) - g(t + \Delta t, t')}{\Delta t} \right]. \quad (15)$$

The Crank-Nicolson discretization becomes:

$$i\hat{\tau}_3[g(t + \Delta t, t') - g(t, t')] + \frac{1}{2} [(g(t, t' + \Delta t) - g(t, t')) + (g(t + \Delta t, t' + \Delta t) - g(t + \Delta t, t'))] i\hat{\tau}_3 = 0. \quad (16)$$

Rearranging to separate knowns (time t) from unknowns (time $t + \Delta t$):

$$\boxed{i\hat{\tau}_3 g(t + \Delta t, t') + \frac{1}{2}(g(t + \Delta t, t' + \Delta t) - g(t + \Delta t, t'))i\hat{\tau}_3 = i\hat{\tau}_3 g(t, t') - \frac{1}{2}(g(t, t' + \Delta t) - g(t, t'))i\hat{\tau}_3.} \quad (17)$$

Key observation: The LHS contains both $g(t + \Delta t, t')$ and $g(t + \Delta t, t' + \Delta t)$. This creates an **implicit system** where all values at the new time $t + \Delta t$ are coupled and must be solved simultaneously.

Adding Unitary Operators

Now consider the equation with unitary operators $\hat{U}(t)$ and $\hat{U}(t')$:

$$i\hat{\tau}_3 \hat{U}(t_1) \frac{\partial g(t_1, t_2)}{\partial t_1} \hat{U}(t_2) + \hat{U}(t_1) \frac{\partial g(t_1, t_2)}{\partial t_2} \hat{U}(t_2) i\hat{\tau}_3 = 0. \quad (18)$$

All objects are matrices.

Goal: Evolve from t to $t + \Delta t$ in the first argument, holding $t_2 = t'$ fixed.

Step 1: Integrate over time interval.

$$\int_t^{t+\Delta t} dt_1 \left[i\hat{\tau}_3 \hat{U}(t_1) \frac{\partial g(t_1, t')}{\partial t_1} \hat{U}(t') + \hat{U}(t_1) \frac{\partial g(t_1, t')}{\partial t'} \hat{U}(t') i\hat{\tau}_3 \right] = 0. \quad (19)$$

Step 2: Crank-Nicolson averaging. Evaluate the integrand at $t_1 = t$ and $t_1 = t + \Delta t$, then average:

At $t_1 = t$:

$$i\hat{\tau}_3 \hat{U}(t) \frac{\partial g(t, t')}{\partial t_1} \Big|_{t_1=t} \hat{U}(t') + \hat{U}(t) \frac{\partial g(t, t')}{\partial t'} \hat{U}(t') i\hat{\tau}_3. \quad (20)$$

At $t_1 = t + \Delta t$:

$$i\hat{\tau}_3 \hat{U}(t + \Delta t) \frac{\partial g(t + \Delta t, t')}{\partial t_1} \Big|_{t_1=t+\Delta t} \hat{U}(t') + \hat{U}(t + \Delta t) \frac{\partial g(t + \Delta t, t')}{\partial t'} \hat{U}(t') i\hat{\tau}_3. \quad (21)$$

Step 3: Rearrange to isolate derivatives. Starting from:

$$i\hat{\tau}_3\hat{U}(t_1)\frac{\partial g(t_1, t')}{\partial t_1}\hat{U}(t') + \hat{U}(t_1)\frac{\partial g(t_1, t')}{\partial t'}\hat{U}(t')i\hat{\tau}_3 = 0. \quad (22)$$

Move the second term to the RHS:

$$i\hat{\tau}_3\hat{U}(t_1)\frac{\partial g(t_1, t')}{\partial t_1}\hat{U}(t') = -\hat{U}(t_1)\frac{\partial g(t_1, t')}{\partial t'}\hat{U}(t')i\hat{\tau}_3. \quad (23)$$

Multiply from the left by $\hat{\tau}_3$ (using $\hat{\tau}_3^2 = 1$):

$$\hat{U}(t_1)\frac{\partial g(t_1, t')}{\partial t_1}\hat{U}(t') = -\hat{\tau}_3\hat{U}(t_1)\frac{\partial g(t_1, t')}{\partial t'}\hat{U}(t')\hat{\tau}_3. \quad (24)$$

Multiply from the left by $\hat{U}^{-1}(t_1)$ and from the right by $\hat{U}^{-1}(t')$:

$$\frac{\partial g(t_1, t')}{\partial t_1} = -\hat{U}^{-1}(t_1)\hat{\tau}_3\hat{U}(t_1)\frac{\partial g(t_1, t')}{\partial t'}\hat{U}(t')\hat{\tau}_3\hat{U}^{-1}(t'). \quad (25)$$

Simplify the right-hand side using $i\hat{\tau}_3 = -\hat{\tau}_3i$:

$$\frac{\partial g(t_1, t')}{\partial t_1} = \hat{U}^{-1}(t_1)\hat{\tau}_3\hat{U}(t_1)\frac{\partial g(t_1, t')}{\partial t'}\hat{U}(t')\hat{\tau}_3\hat{U}^{-1}(t'). \quad (26)$$

Step 4: Integrate over $[t, t + \Delta t]$.

$$\int_t^{t+\Delta t} dt_1 \frac{\partial g(t_1, t')}{\partial t_1} = \int_t^{t+\Delta t} dt_1 \hat{U}^{-1}(t_1)\hat{\tau}_3\hat{U}(t_1)\frac{\partial g(t_1, t')}{\partial t'}\hat{U}(t')\hat{\tau}_3\hat{U}^{-1}(t'). \quad (27)$$

Left-hand side:

$$\text{LHS} = g(t + \Delta t, t') - g(t, t'). \quad (28)$$

Right-hand side (Crank-Nicolson): evaluate at $t_1 = t$ and $t_1 = t + \Delta t$, then average and multiply by Δt :

$$\begin{aligned} \text{RHS} = \frac{\Delta t}{2} & \left[\hat{U}^{-1}(t)\hat{\tau}_3\hat{U}(t)\frac{\partial g(t, t')}{\partial t'}\hat{U}(t')\hat{\tau}_3\hat{U}^{-1}(t') \right. \\ & \left. + \hat{U}^{-1}(t + \Delta t)\hat{\tau}_3\hat{U}(t + \Delta t)\frac{\partial g(t + \Delta t, t')}{\partial t'}\hat{U}(t')\hat{\tau}_3\hat{U}^{-1}(t') \right]. \end{aligned} \quad (29)$$

Step 5: Discretize the $\partial_{t'}$ derivatives. Using finite differences:

$$\frac{\partial g(t, t')}{\partial t'} \approx \frac{g(t, t' + \Delta t) - g(t, t')}{\Delta t}, \quad (30)$$

$$\frac{\partial g(t + \Delta t, t')}{\partial t'} \approx \frac{g(t + \Delta t, t' + \Delta t) - g(t + \Delta t, t')}{\Delta t}. \quad (31)$$

Substituting:

$$\begin{aligned} g(t + \Delta t, t') - g(t, t') &= \frac{1}{2} \left[\hat{U}^{-1}(t) \hat{\tau}_3 \hat{U}(t) [g(t, t' + \Delta t) - g(t, t')] \hat{U}(t') \hat{\tau}_3 \hat{U}^{-1}(t') \right. \\ &\quad + \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t) [g(t + \Delta t, t' + \Delta t) \\ &\quad \left. - g(t + \Delta t, t')] \hat{U}(t') \hat{\tau}_3 \hat{U}^{-1}(t') \right]. \end{aligned} \quad (32)$$

Step 6: Multiply by unitary operators. Multiply the entire equation from the left by $\hat{U}(t)$ and from the right by $\hat{U}(t')$:

$$\begin{aligned} \hat{U}(t) [g(t + \Delta t, t') - g(t, t')] \hat{U}(t') &= \frac{1}{2} \left[\hat{U}(t) \hat{U}^{-1}(t) \hat{\tau}_3 \hat{U}(t) [g(t, t' + \Delta t) - g(t, t')] \hat{U}(t') \hat{\tau}_3 \hat{U}^{-1}(t') \hat{U}(t') \right. \\ &\quad + \hat{U}(t) \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t) [g(t + \Delta t, t' + \Delta t) \\ &\quad \left. - g(t + \Delta t, t')] \hat{U}(t') \hat{\tau}_3 \hat{U}^{-1}(t') \hat{U}(t') \right]. \end{aligned} \quad (33)$$

Simplify using $\hat{U} \hat{U}^{-1} = 1$:

$$\begin{aligned} \hat{U}(t) g(t + \Delta t, t') \hat{U}(t') - \hat{U}(t) g(t, t') \hat{U}(t') &= \frac{1}{2} \left[\hat{\tau}_3 \hat{U}(t) [g(t, t' + \Delta t) - g(t, t')] \hat{U}(t') \hat{\tau}_3 \right. \\ &\quad + \hat{U}(t) \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t) [g(t + \Delta t, t' + \Delta t) \\ &\quad \left. - g(t + \Delta t, t')] \hat{U}(t') \hat{\tau}_3 \right]. \end{aligned} \quad (34)$$

Step 7: Separate knowns from unknowns. Expand the LHS:

$$\hat{U}(t) g(t + \Delta t, t') \hat{U}(t') = \hat{U}(t) g(t, t') \hat{U}(t') + \text{RHS}. \quad (35)$$

Expand the RHS terms involving $g(t + \Delta t, \cdot)$:

$$\begin{aligned} \text{RHS} &= \frac{1}{2} \hat{\tau}_3 \hat{U}(t) [g(t, t' + \Delta t) - g(t, t')] \hat{U}(t') \hat{\tau}_3 \\ &\quad + \frac{1}{2} \hat{U}(t) \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t) g(t + \Delta t, t' + \Delta t) \hat{U}(t') \hat{\tau}_3 \\ &\quad - \frac{1}{2} \hat{U}(t) \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t) g(t + \Delta t, t') \hat{U}(t') \hat{\tau}_3. \end{aligned} \quad (36)$$

Move the last term (containing $g(t + \Delta t, t')$) to the LHS:

$$\begin{aligned} \hat{U}(t) g(t + \Delta t, t') \hat{U}(t') &+ \frac{1}{2} \hat{U}(t) \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t) g(t + \Delta t, t') \hat{U}(t') \hat{\tau}_3 \\ &= \hat{U}(t) g(t, t') \hat{U}(t') + \frac{1}{2} \hat{\tau}_3 \hat{U}(t) [g(t, t' + \Delta t) - g(t, t')] \hat{U}(t') \hat{\tau}_3 \\ &\quad + \frac{1}{2} \hat{U}(t) \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t) g(t + \Delta t, t' + \Delta t) \hat{U}(t') \hat{\tau}_3. \end{aligned} \quad (37)$$

Step 8: Factor out common terms. On the LHS, factor using the identity:

$$A + \frac{1}{2} B \hat{\tau}_3 A \hat{\tau}_3 = \left[1 + \frac{1}{2} B \hat{\tau}_3 \cdot \hat{\tau}_3 \right] A, \quad (38)$$

where $A = \hat{U}(t) g(t + \Delta t, t') \hat{U}(t')$ and $B = \hat{U}(t) \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t)$.

Actually, this doesn't factor cleanly. Let me write the final form directly:

LHS (unknowns at time $t + \Delta t$):

$$\begin{aligned} &\hat{U}(t) g(t + \Delta t, t') \hat{U}(t') + \frac{1}{2} \hat{U}(t) \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t) g(t + \Delta t, t') \hat{U}(t') \hat{\tau}_3 \\ &- \frac{1}{2} \hat{U}(t) \hat{U}^{-1}(t + \Delta t) \hat{\tau}_3 \hat{U}(t + \Delta t) g(t + \Delta t, t' + \Delta t) \hat{U}(t') \hat{\tau}_3 \end{aligned}$$

(39)

RHS (knowns at time t):

$$\hat{U}(t) g(t, t') \hat{U}(t') + \frac{1}{2} \hat{\tau}_3 \hat{U}(t) [g(t, t' + \Delta t) - g(t, t')] \hat{U}(t') \hat{\tau}_3$$

(40)

Step 9: Simplify unitary operator products. Using the composition property $\hat{U}(t + \Delta t) = \hat{U}(\Delta t; t) \hat{U}(t)$, we have:

$$\hat{U}^{-1}(t + \Delta t) = \hat{U}^{-1}(t) \hat{U}^{-1}(\Delta t; t), \quad (41)$$

$$\hat{U}(t) \hat{U}^{-1}(t + \Delta t) = \hat{U}(t) \hat{U}^{-1}(t) \hat{U}^{-1}(\Delta t; t) = \hat{U}^{-1}(\Delta t; t). \quad (42)$$

Substituting into the LHS:

$$\begin{aligned} \text{LHS} &= \hat{U}(t)g(t + \Delta t, t')\hat{U}(t') + \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(t + \Delta t)g(t + \Delta t, t')\hat{U}(t')\hat{\tau}_3 \\ &\quad - \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(t + \Delta t)g(t + \Delta t, t' + \Delta t)\hat{U}(t')\hat{\tau}_3. \end{aligned} \quad (43)$$

Now use $\hat{U}(t + \Delta t) = \hat{U}(\Delta t; t)\hat{U}(t)$:

$$\begin{aligned} \text{LHS} &= \hat{U}(t)g(t + \Delta t, t')\hat{U}(t') + \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)\hat{U}(t)g(t + \Delta t, t')\hat{U}(t')\hat{\tau}_3 \\ &\quad - \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)\hat{U}(t)g(t + \Delta t, t' + \Delta t)\hat{U}(t')\hat{\tau}_3. \end{aligned} \quad (44)$$

Factor out $\hat{U}(t)(\dots)\hat{U}(t')$ from each term:

$$\begin{aligned} \text{LHS} &= \hat{U}(t) \left[g(t + \Delta t, t') + \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)g(t + \Delta t, t')\hat{\tau}_3 \right. \\ &\quad \left. - \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)g(t + \Delta t, t' + \Delta t)\hat{\tau}_3 \right] \hat{U}(t'). \end{aligned} \quad (45)$$

Wait, this doesn't work because the $\hat{\tau}_3$ on the right doesn't commute with $\hat{U}(t')$. Let me reconsider.

Actually, for the RHS we also need to expand similarly. Let me also use $\hat{U}(t') = \hat{U}(\Delta t; t')\hat{U}(t' - \Delta t)$ for the terms involving $\hat{U}(t')$.

But actually, t' is held fixed, so we can't decompose $\hat{U}(t')$ in the same way unless we also consider the evolution in the t' direction.

Let me simplify differently. Define the relative unitary propagator:

$$\hat{U}_R^{-1}(\Delta t; t') \equiv \hat{U}(t')\hat{U}^{-1}(t' + \Delta t) = \hat{U}(t')\hat{U}^{-1}(t')\hat{U}^{-1}(\Delta t; t') = \hat{U}^{-1}(\Delta t; t'). \quad (46)$$

So the simplified form is:

LHS (simplified):

$$\begin{aligned} &\hat{U}(t)g(t + \Delta t, t')\hat{U}(t') \\ &+ \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)\hat{U}(t)g(t + \Delta t, t')\hat{U}(t')\hat{\tau}_3 \\ &- \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)\hat{U}(t)g(t + \Delta t, t' + \Delta t)\hat{U}(t')\hat{\tau}_3 \end{aligned}$$

(47)

For the term $g(t + \Delta t, t' + \Delta t)\hat{U}(t')$, we can write:

$$g(t + \Delta t, t' + \Delta t)\hat{U}(t') = g(t + \Delta t, t' + \Delta t)\hat{U}(t' + \Delta t)\hat{U}^{-1}(\Delta t; t'). \quad (48)$$

So:

$$\begin{aligned} & \hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)\hat{U}(t)g(t + \Delta t, t' + \Delta t)\hat{U}(t')\hat{\tau}_3 \\ &= \hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)\hat{U}(t)g(t + \Delta t, t' + \Delta t)\hat{U}(t' + \Delta t)\hat{U}^{-1}(\Delta t; t')\hat{\tau}_3. \end{aligned} \quad (49)$$

Final simplified form:

LHS:

$$\hat{U}(t) \left[g(t + \Delta t, t') + \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)g(t + \Delta t, t')\hat{\tau}_3 - \frac{1}{2}\hat{U}^{-1}(\Delta t; t)\hat{\tau}_3\hat{U}(\Delta t; t)g(t + \Delta t, t' + \Delta t)\hat{U}^{-1}(\Delta t; t')\hat{\tau}_3 \right] \hat{U}(t'). \quad (50)$$

RHS:

$$\hat{U}(t) \left[g(t, t') + \frac{1}{2}\hat{\tau}_3[g(t, t' + \Delta t) - g(t, t')]\hat{\tau}_3 \right] \hat{U}(t'). \quad (51)$$